

FES SAIS, Morocco

WMO: 601410

Lat: 33.927N

Lon: 4.978W

Elev: 1900

StdP: 13.71

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.7	35.8	22.9	18.5	47.1	25.6	21.0	48.4	29.2	52.0	24.8	52.6	6.3	180	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	29.1	104.1	67.4	100.5	67.5	96.9	67.2	71.7	93.0	70.5	91.6	69.3	89.9	10.6	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.3	100.3	78.1	64.2	96.4	77.4	62.7	91.5	76.2	36.6	92.4	35.4	91.1	34.4	90.2	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.0	19.0	16.0	DB	29.0	108.9	3.1	2.5	26.8	110.6	25.0	112.1	23.2	113.5	20.9	115.3	
			WB	26.7	74.7	3.0	2.4	24.5	76.4	22.8	77.8	21.1	79.2	18.9	80.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.2	50.3	51.8	56.2	59.4	66.3	74.1	79.7	80.4	73.4	67.6	57.8	52.3
	DBStd	11.89	4.28	4.61	4.92	5.17	6.90	6.58	6.20	5.85	5.47	5.72	5.87	4.43
	HDD50	119	49	31	7	1	0	0	0	0	0	0	5	26
	HDD65	2021	457	371	276	182	67	3	0	0	4	38	228	395
	CDD50	5292	57	80	199	283	506	722	921	943	703	544	238	96
	CDD65	1717	0	0	2	13	108	275	456	478	257	117	11	0
	CDH74	22262	1	6	101	321	1547	3601	6204	6331	2766	1251	127	6
	CDH80	11940	0	1	14	76	697	1916	3722	3760	1290	450	14	0
Wind	WSAvg	7.4	7.6	7.9	7.8	7.8	7.4	7.5	7.1	7.2	7.0	6.9	7.7	7.6
Precipitation	PrecAvg	20.9	2.4	2.8	3.0	2.6	1.5	0.8	0.1	0.1	0.5	1.8	2.5	2.6
	PrecMax	31.7	7.6	7.6	6.0	6.7	5.0	2.6	0.4	0.7	2.4	5.6	6.7	6.9
	PrecMin	11.0	0.0	0.0	0.4	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	5.4	2.1	2.1	1.7	1.9	1.3	0.7	0.1	0.2	0.6	1.8	1.8	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.8	73.8	81.8	86.4	97.2	105.4	109.0	107.7	100.7	93.2	82.2	73.8
		MCWB	54.0	54.4	60.9	62.0	64.4	66.7	67.5	67.1	65.8	62.2	58.8	56.1
	2%	DB	67.6	69.9	76.7	82.0	92.6	100.0	104.2	103.9	95.3	89.0	77.4	68.6
		MCWB	53.0	54.5	58.6	61.0	64.0	66.3	67.9	67.4	65.8	62.2	57.5	53.9
	5%	DB	64.3	66.5	73.1	77.3	87.9	95.1	100.4	100.4	91.3	85.5	73.8	65.8
		MCWB	52.0	54.3	57.5	59.5	63.1	66.1	67.7	68.2	65.7	61.9	57.3	53.6
	10%	DB	60.9	63.3	69.5	73.3	82.7	91.0	96.6	96.6	87.6	81.4	70.0	62.6
		MCWB	51.4	53.1	56.4	59.2	62.3	65.4	67.7	67.8	65.4	61.6	56.7	52.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.9	59.1	64.4	66.6	68.4	71.4	73.3	73.7	70.9	67.3	65.3	60.1
		MCDB	65.1	67.4	75.5	78.7	86.9	92.5	94.9	94.9	88.8	81.7	72.3	67.4
	2%	WB	55.6	57.2	61.5	63.6	66.4	69.5	71.6	71.9	69.1	65.7	61.5	57.4
		MCDB	63.1	65.5	71.9	76.4	84.5	90.4	93.7	93.1	86.0	79.1	71.4	63.9
	5%	WB	53.8	55.5	59.3	61.5	64.8	68.2	70.3	70.6	67.9	64.4	59.3	55.4
		MCDB	60.8	63.7	69.5	73.9	82.5	88.5	92.7	91.7	84.0	77.2	69.1	62.2
	10%	WB	52.1	53.9	57.4	59.5	63.2	66.8	69.0	69.3	66.7	63.2	57.5	53.7
		MCDB	59.3	61.7	66.7	70.6	79.5	86.2	91.1	90.1	82.6	75.9	67.0	60.6
Mean Daily Temperature Range	5% DB	MDBR	20.4	21.6	22.9	23.3	25.3	28.0	29.6	29.1	25.2	23.4	20.5	20.2
		MCDBR	27.2	28.1	30.6	32.3	34.6	34.0	33.1	33.2	31.4	30.0	26.6	25.3
		MCWBR	16.4	16.8	17.0	15.7	13.5	12.0	11.7	12.0	12.0	12.9	13.6	14.9
	5% WB	MCDBR	20.8	24.0	25.9	27.4	29.7	30.1	30.0	29.3	25.7	23.7	21.1	20.0
		MCWBR	13.7	15.3	15.9	14.9	12.8	11.8	12.0	11.7	10.7	11.2	11.7	13.0
Clear-Sky Solar Irradiance	taub	0.294	0.329	0.382	0.395	0.428	0.464	0.511	0.486	0.430	0.387	0.333	0.303	
	taud	2.540	2.434	2.266	2.251	2.179	2.069	1.927	2.005	2.191	2.315	2.472	2.540	
	Ebn at Noon	294	292	282	283	273	262	249	253	263	266	275	284	
	Edh at Noon	26	32	41	43	47	53	60	55	43	36	27	24	
All-Sky Solar Radiation	RadAvg	953	1222	1595	1903	2159	2442	2422	2182	1785	1349	1002	880	
	RadStd	103	126	126	138	158	96	60	74	93	98	90	81	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.92	N/A	N/A	+2.05	+0.67	N/A	N/A	-158	+508	+382
(47)	Regional (6 neighbors)	N/A	N/A	N/A	+1.89	N/A	-0.86	N/A	N/A	N/A	+146

Nomenclature: See separate page